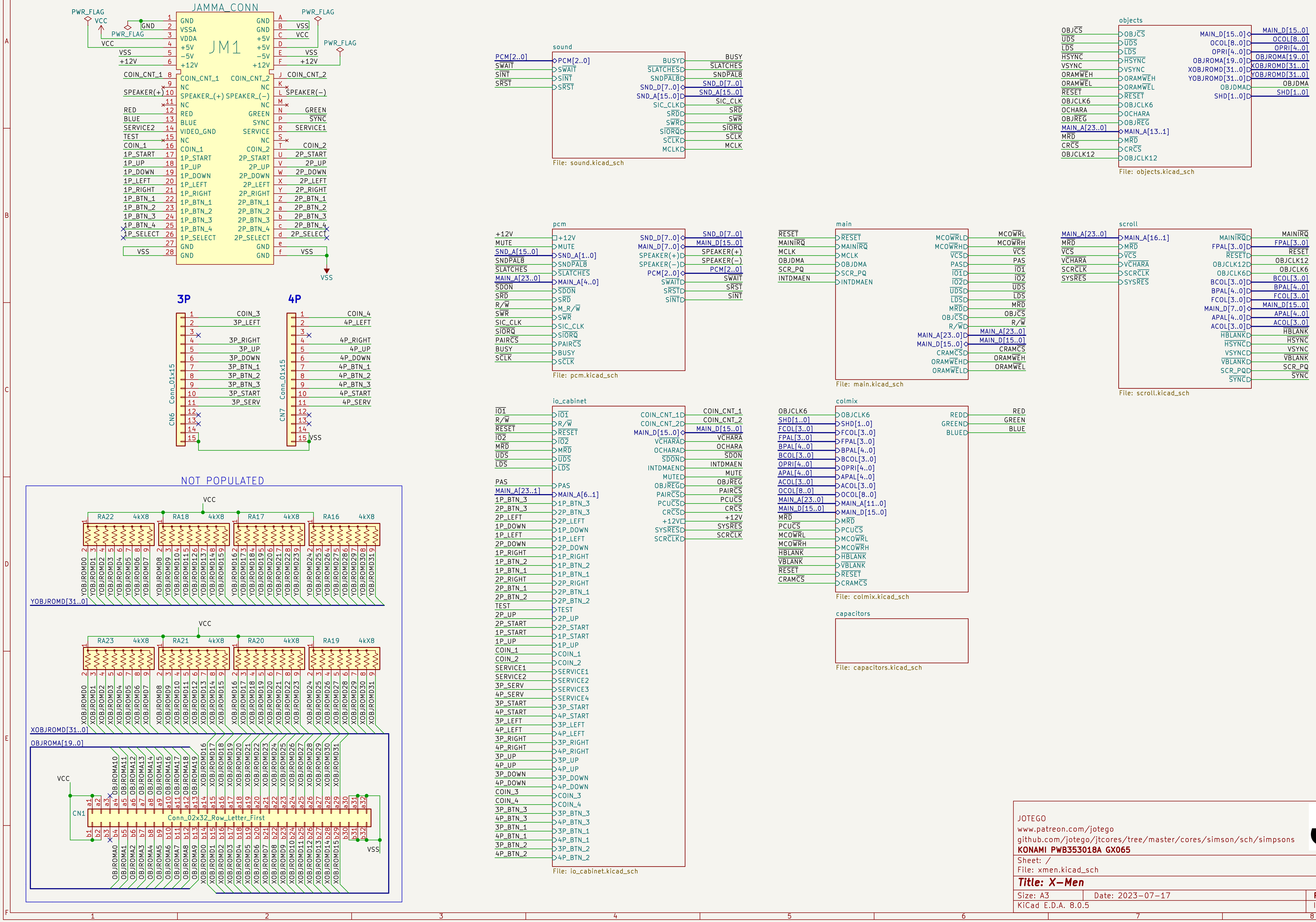
JAMMA_CONN
JM1
PWR_FLAG
VCC
VSS
+12V
GND
3P
4P
CN6 Conn_01x15
CN7 Conn_01x15
NOT POPULATED
RA22 4kX8
RA18 4kX8
RA17 4kX8
RA16 4kX8
RA23 4kX8
RA21 4kX8
RA20 4kX8
RA19 4kX8
YOBJROMD[31..0]
XOBJROMD[31..0]
OBJROMA[19..0]
CN1 Conn_02x32_Row_Letter_First
sound
File: sound.kicad_sch
pcm
File: pcm.kicad_sch
io_cabinet
File: io_cabinet.kicad_sch
objects
File: objects.kicad_sch
main
File: main.kicad_sch
scroll
File: scroll.kicad_sch
colmix
File: colmix.kicad_sch
capacitors
File: capacitors.kicad_sch

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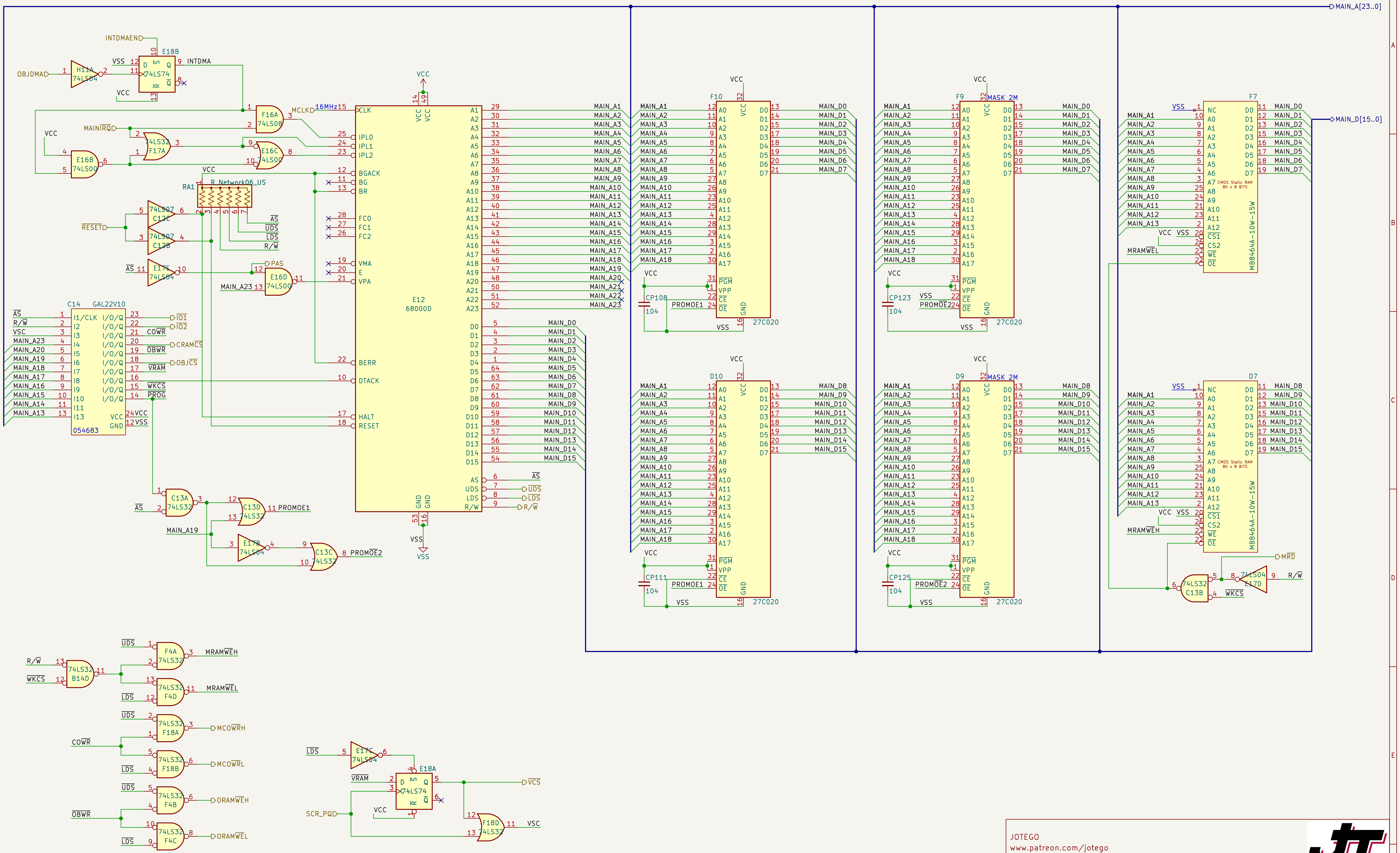
KONAMI PWB353018A GX065

Sheet: /
File: xmen.kicad_sch

Title: X-Men

Size: A3	Date: 2023-07-17	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 1/10


E12
68000D
F10
27C020
D10
F9
D9
MASK 2M
F7
D7
MB8464A-10W-15W
CMOS Static RAM 8K x 8 BITS
C14
GAL22V10
054683
RA1
R_Network06_US
E18B
E18A
74LS74
74LS04
74LS08
74LS00
74LS32
74LS07
H11A
F16A
F17A
E16B
E16C
C12C
C12B
E17F
E16D
C13A
C13D
E17B
C13C
C13B
E17D
B14D
F4A
F4D
F18A
F18B
F4B
F4C
E17C
F18D
CP108
CP123
CP111
CP125
104
MAIN_A[23..0]
MAIN_D[15..0]
OBJDMA
INTDMAEN
INTDMA
MAINIRQ
MCLK
16MHz
RESET
PAS
PROMOE1
PROMOE2
MRAMWEH
MRAMWEL
MCOWRH
MCOWRL
ORAMWEH
ORAMWEL
COWR
OBWR
CRAMCS
OBJCS
VRAM
WKCS
PROG
IO1
IO2
VSC
VCS
MRD
SCR_PQ
AS
UDS
LDS
R/W
VCC
VSS

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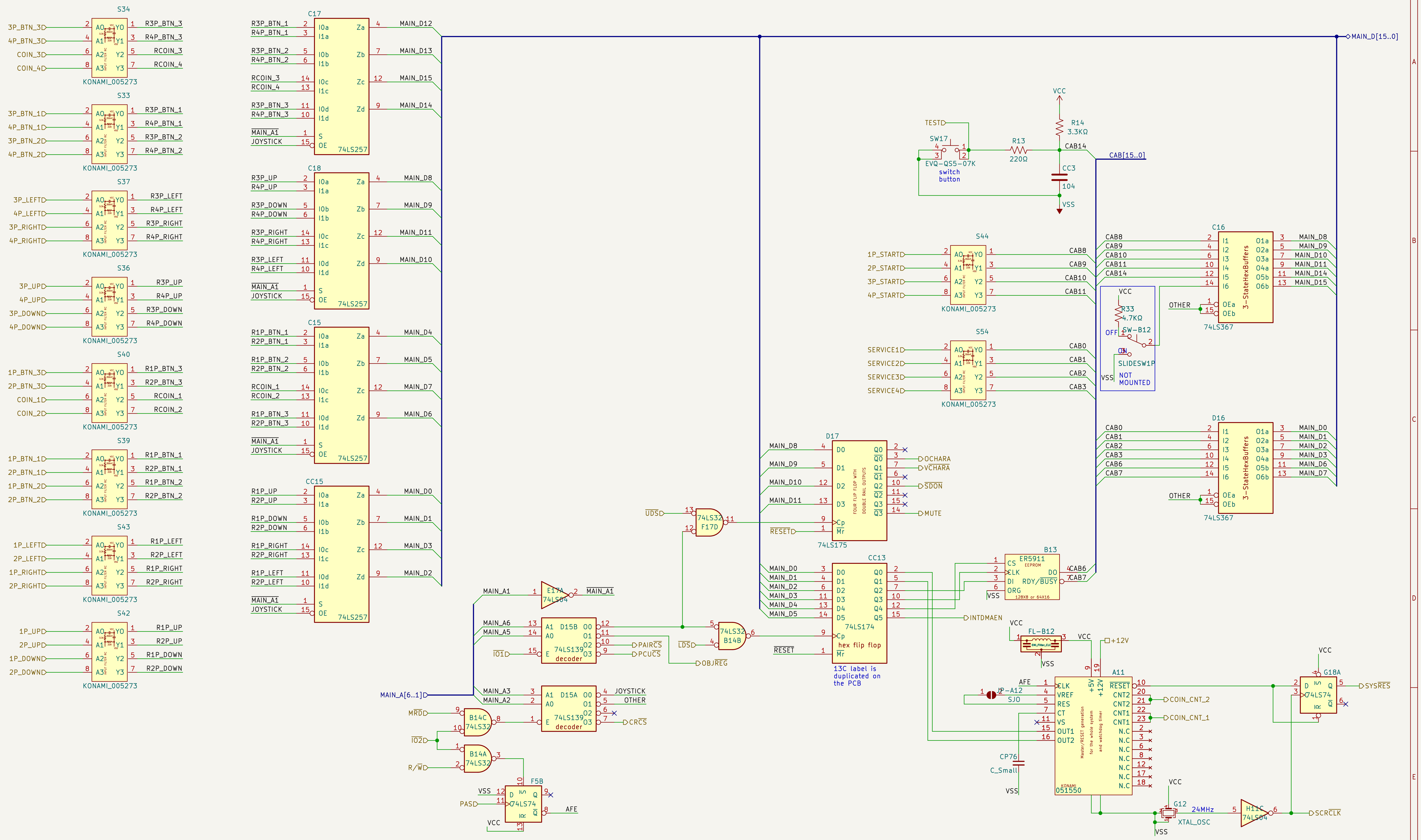
Sheet: /main/
File: main.kicad_sch

Title: X–Men

Size: A3 Date: 2023-07-17 **Rev: José Tejada**
KiCad E.D.A. 8.0.5 Id: 2/10


S34
3P_BTN_3
4P_BTN_3
COIN_3
COIN_4
R3P_BTN_3
R4P_BTN_3
RCOIN_3
RCOIN_4
KONAMI_005273
S33
3P_BTN_1
4P_BTN_1
3P_BTN_2
4P_BTN_2
R3P_BTN_1
R4P_BTN_1
R3P_BTN_2
R4P_BTN_2
KONAMI_005273
S37
3P_LEFT
4P_LEFT
3P_RIGHT
4P_RIGHT
R3P_LEFT
R4P_LEFT
R3P_RIGHT
R4P_RIGHT
KONAMI_005273
S36
3P_UP
4P_UP
3P_DOWN
4P_DOWN
R3P_UP
R4P_UP
R3P_DOWN
R4P_DOWN
KONAMI_005273
S40
1P_BTN_3
2P_BTN_3
COIN_1
COIN_2
R1P_BTN_3
R2P_BTN_3
RCOIN_1
RCOIN_2
KONAMI_005273
S39
1P_BTN_1
2P_BTN_1
1P_BTN_2
2P_BTN_2
R1P_BTN_1
R2P_BTN_1
R1P_BTN_2
R2P_BTN_2
KONAMI_005273
S43
1P_LEFT
2P_LEFT
1P_RIGHT
2P_RIGHT
R1P_LEFT
R2P_LEFT
R1P_RIGHT
R2P_RIGHT
KONAMI_005273
S42
1P_UP
2P_UP
1P_DOWN
2P_DOWN
R1P_UP
R2P_UP
R1P_DOWN
R2P_DOWN
KONAMI_005273
C17
74LS257
MAIN_D12
MAIN_D13
MAIN_D15
MAIN_D14
MAIN_A1
JOYSTICK
C18
74LS257
MAIN_D8
MAIN_D9
MAIN_D11
MAIN_D10
C15
74LS257
MAIN_D4
MAIN_D5
MAIN_D7
MAIN_D6
CC15
74LS257
MAIN_D0
MAIN_D1
MAIN_D3
MAIN_D2
MAIN_D[15..0]
VCC
TEST
SW17
EVQ-QS5-07K
switch
button
R13
220Ω
R14
3.3KΩ
CAB14
CC3
104
VSS
CAB[15..0]
S44
1P_START
2P_START
3P_START
4P_START
CAB8
CAB9
CAB10
CAB11
KONAMI_005273
S54
SERVICE1
SERVICE2
SERVICE3
SERVICE4
CAB0
CAB1
CAB2
CAB3
KONAMI_005273
R33
4.7KΩ
SW-B12
OFF
ON
SLIDESW1P
NOT
MOUNTED
C16
3-StateHexBuffers
74LS367
CAB8
CAB9
CAB10
CAB11
CAB14
MAIN_D8
MAIN_D9
MAIN_D10
MAIN_D11
MAIN_D14
MAIN_D15
OTHER
D16
3-StateHexBuffers
74LS367
CAB0
CAB1
CAB2
CAB3
CAB6
CAB7
MAIN_D0
MAIN_D1
MAIN_D2
MAIN_D3
MAIN_D6
MAIN_D7
D17
74LS175
FOUR FLIP FLOP WITH DOUBLE RAIL OUTPUTS
OCHARA
VCHARA
SDON
MUTE
UDS
74LS32
F17D
RESET
CC13
74LS174
hex flip flop
13C label is duplicated on the PCB
INTDMAEN
B13
ER5911
EEPROM
128X8 or 64X16
CAB6
CAB7
E17A
74LS04
MAIN_A1
D15B
74LS139
decoder
MAIN_A6
MAIN_A5
IOI
PAIRCS
PCUCS
OBJREG
B14B
74LS32
LDS
MAIN_A[6..1]
D15A
74LS139
decoder
MAIN_A3
MAIN_A2
JOYSTICK
OTHER
CRCS
MRD
B14C
74LS32
IO2
B14A
74LS32
R/W
F5B
74LS74
VSS
PASD
VCC
AFE
FL-B12
+12V
JP-A12
SJ0
A11
051550
KONAMI
Master/RESET generation for the whole system and watchdog timer
COIN_CNT_2
COIN_CNT_1
CP76
C_Small
G12
XTAL_OSC
24MHz
H11C
74LS04
SCRCLK
G18A
74LS74
SYSRES

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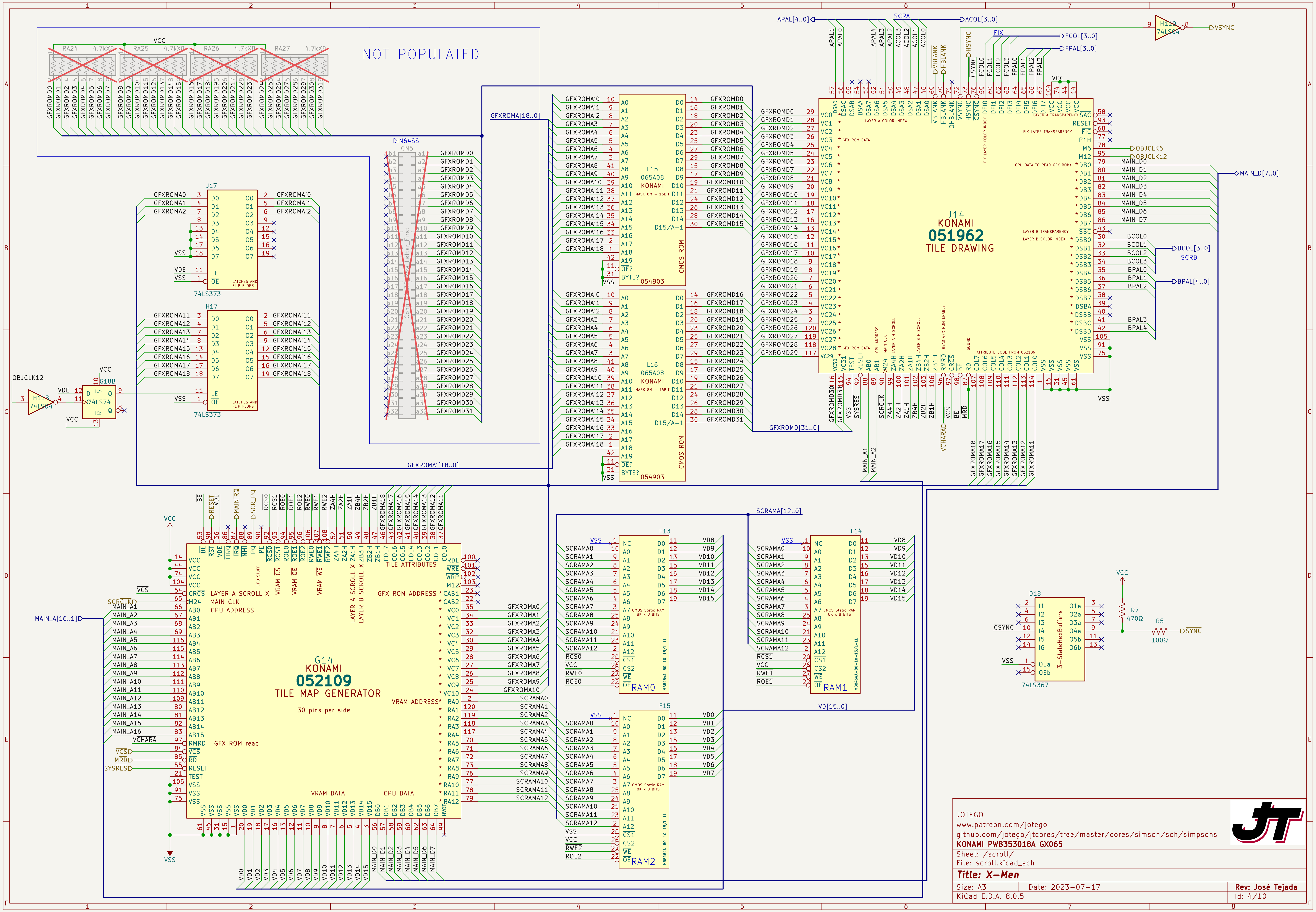
KONAMI PWB353018A GX065
Sheet: /io_cabinet/
File: io_cabinet.kicad_sch

Title: X-Men

Size: A3
Date: 2023-07-17
Rev: José Tejada
KiCad E.D.A. 8.0.5
Id: 3/10


NOT POPULATED
RA24 4.7kX8
RA25 4.7kX8
RA26 4.7kX8
RA27 4.7kX8
VCC
DIN64SS
CN5
GFXROMD[31..0]
GFXROMA[18..0]
GFXROMA'[18..0]
J17
74LS373
H17
74LS373
LATCHES AND FLIP FLOPS
OBJCLK12
H11B
74LS04
G18B
74LS74
VDE
VSS
L15
065A08
KONAMI
MASK 8M - 16BIT
CMOS ROM
054903
L16
065A08
KONAMI
APAL[4..0]
SCRA
ACOL[3..0]
FIX
FCOL[3..0]
FPAL[3..0]
H11D
74LS04
VSYNC
J14
KONAMI
051962
TILE DRAWING
LAYER A COLOR INDEX
LAYER A TRANSPARENCY
FIX LAYER TRANSPARENCY
FIX LAYER COLOR INDEX
GFX ROM DATA
CPU DATA TO READ GFX ROMs
LAYER B TRANSPARENCY
LAYER B COLOR INDEX
ATTRIBUTE CODE FROM 052109
OBJCLK6
OBJCLK12
MAIN_D[7..0]
BCOL[3..0]
SCRB
BPAL[4..0]
VCHARA
MAIN_A1
MAIN_A2
SCRAMA[12..0]
G14
KONAMI
052109
TILE MAP GENERATOR
30 pins per side
LAYER A SCROLL X
MAIN CLK
CPU ADDRESS
TILE ATTRIBUTES
GFX ROM ADDRESS
VRAM ADDRESS
GFX ROM read
VRAM DATA
CPU DATA
LAYER A SCROLL X ZA1H
LAYER B SCROLL X ZB3H
MAIN_A[16..1]
SCRCLK
VCSD
MRD
SYSRES
F13
RAM0
F14
RAM1
F15
RAM2
CMOS Static RAM 8K x 8 BITS
MB8464A-80-10-15/L-LL
VD[15..0]
D18
74LS367
3-StateHexBuffers
CSYNC
R7
470Ω
R5
100Ω
SYNC

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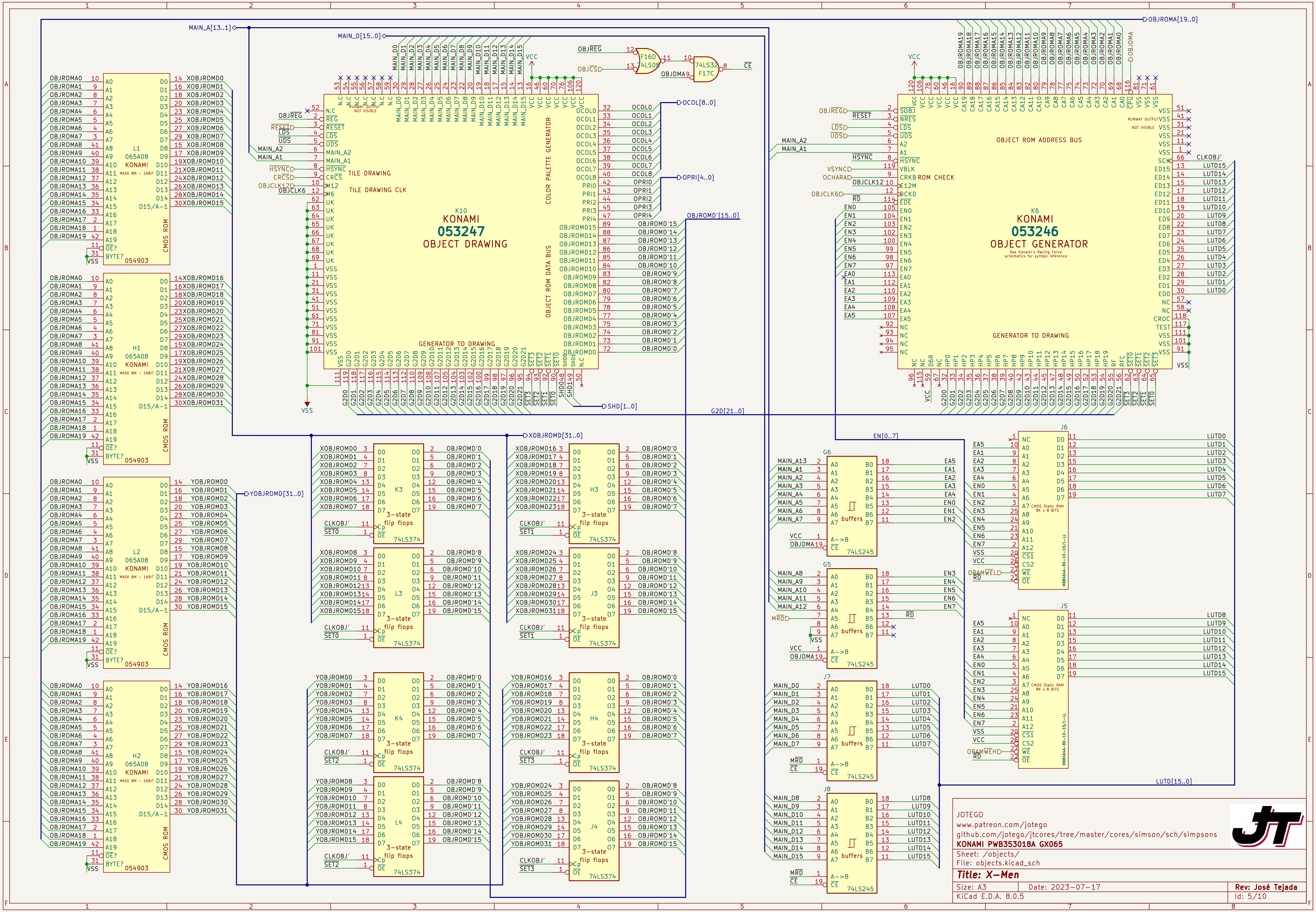
KONAMI PWB353018A GX065
Sheet: /scroll/
File: scroll.kicad_sch

Title: X–Men

Size: A3 Date: 2023-07-17 **Rev: José Tejada**
KiCad E.D.A. 8.0.5 Id: 4/10


MAIN_A[13..1]
MAIN_D[15..0]
OBJROMA[19..0]
OBJREG
OBJCS
F16D 74LS08
74LS32 F17C
OBJDMA9
CE
OCOL[8..0]
OPRI[4..0]
OBJROMD'[15..0]
SHD[1..0]
G2D[21..0]
XOBJROMD[31..0]
YOBJROMD[31..0]
EN[0..7]
LUTD[15..0]
L1 065A08 KONAMI MASK 8M - 16BIT CMOS ROM 054903
H1 065A08 KONAMI MASK 8M - 16BIT CMOS ROM 054903
L2 065A08 KONAMI MASK 8M - 16BIT CMOS ROM 054903
H2 065A08 KONAMI MASK 8M - 16BIT CMOS ROM 054903
K10 KONAMI 053247 OBJECT DRAWING
TILE DRAWING
TILE DRAWING CLK
COLOR PALETTE GENERATOR
OBJECT ROM DATA BUS
GENERATOR TO DRAWING
RUNWAY OUTPUT NOT VISIBLE
K6 KONAMI 053246 OBJECT GENERATOR
See Konami's Racing Force schematics for symbol reference
OBJECT ROM ADDRESS BUS
ROM CHECK
GENERATOR TO DRAWING
K3 3-state flip flops 74LS374
H3 3-state flip flops 74LS374
L3 3-state flip flops 74LS374
J3 3-state flip flops 74LS374
K4 3-state flip flops 74LS374
H4 3-state flip flops 74LS374
L4 3-state flip flops 74LS374
J4 3-state flip flops 74LS374
G6 buffers 74LS245
G5 buffers 74LS245
J7 buffers 74LS245
J8 buffers 74LS245
J6 CMOS Static RAM 8K x 8 BITS MB8464A-80-10-15/L-LL
J5 CMOS Static RAM 8K x 8 BITS MB8464A-80-10-15/L-LL
OBJRAMWEL
OBJRAMWEH
CLKOBJ'
VCC
VSS

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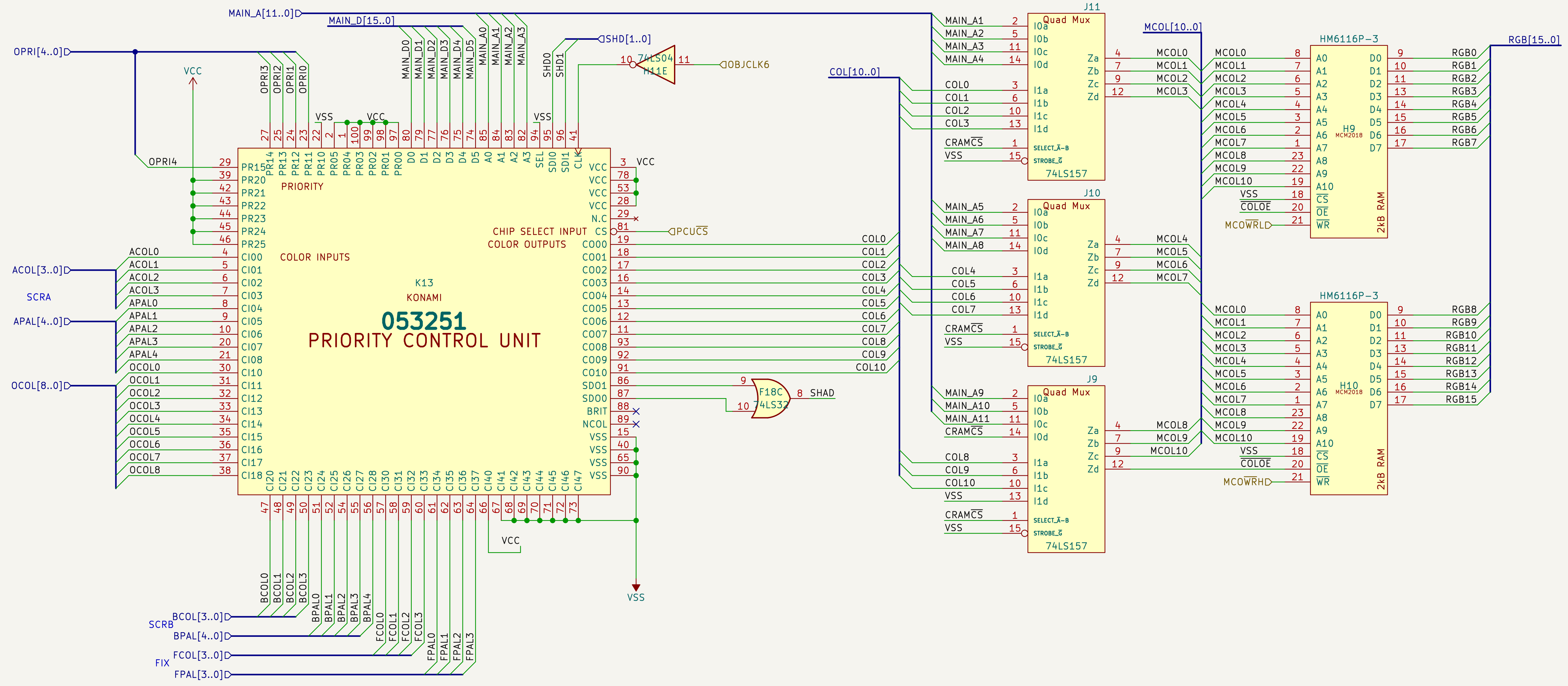
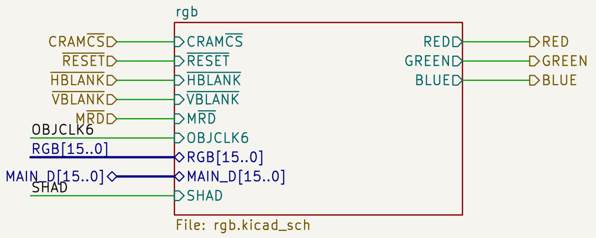
KONAMI PWB353018A GX065
Sheet: /objects/
File: objects.kicad_sch

Title: X-Men

Size: A3 Date: 2023-07-17 **Rev: José Tejada**
KiCad E.D.A. 8.0.5 Id: 5/10



MAIN_A[11..0]
MAIN_D[15..0]
OPRI[4..0]
OPRI4
OPRI3
OPRI2
OPRI1
OPRI0
VCC
VSS
SHD[1..0]
SHD0
SHD1
74LS04
H11E
OBJCLK6
ACOL[3..0]
SCRA
APAL[4..0]
OCOL[8..0]
ACOL0
ACOL1
ACOL2
ACOL3
APAL0
APAL1
APAL2
APAL3
APAL4
OCOL0
OCOL1
OCOL2
OCOL3
OCOL4
OCOL5
OCOL6
OCOL7
OCOL8
PRIORITY
COLOR INPUTS
CHIP SELECT INPUT
COLOR OUTPUTS
K13
KONAMI
053251
PRIORITY CONTROL UNIT
PCUCS
BRIT
NCOL
SDO1
SDO0
F18C
74LS32
SHAD
COL[10..0]
COL0
COL1
COL2
COL3
COL4
COL5
COL6
COL7
COL8
COL9
COL10
BCOL[3..0]
SCRB
BPAL[4..0]
FCOL[3..0]
FIX
FPAL[3..0]
BCOL0
BCOL1
BCOL2
BCOL3
BPAL0
BPAL1
BPAL2
BPAL3
BPAL4
FCOL0
FCOL1
FCOL2
FCOL3
FPAL0
FPAL1
FPAL2
FPAL3
J11
J10
J9
Quad Mux
74LS157
MAIN_A1
MAIN_A2
MAIN_A3
MAIN_A4
MAIN_A5
MAIN_A6
MAIN_A7
MAIN_A8
MAIN_A9
MAIN_A10
MAIN_A11
CRAMCS
SELECT_A-B
STROBE_G
MCOL[10..0]
MCOL0
MCOL1
MCOL2
MCOL3
MCOL4
MCOL5
MCOL6
MCOL7
MCOL8
MCOL9
MCOL10
HM6116P-3
H9
H10
MCM2018
2kB RAM
COLOE
MCOWRL
MCOWRH
RGB[15..0]
RGB0
RGB1
RGB2
RGB3
RGB4
RGB5
RGB6
RGB7
RGB8
RGB9
RGB10
RGB11
RGB12
RGB13
RGB14
RGB15



rgb
CRAMCS
RESET
HBLANK
VBLANK
MRD
OBJCLK6
RGB[15..0]
MAIN_D[15..0]
SHAD
REDD
GREEND
BLUED
RED
GREEN
BLUE
File: rgb.kicad_sch

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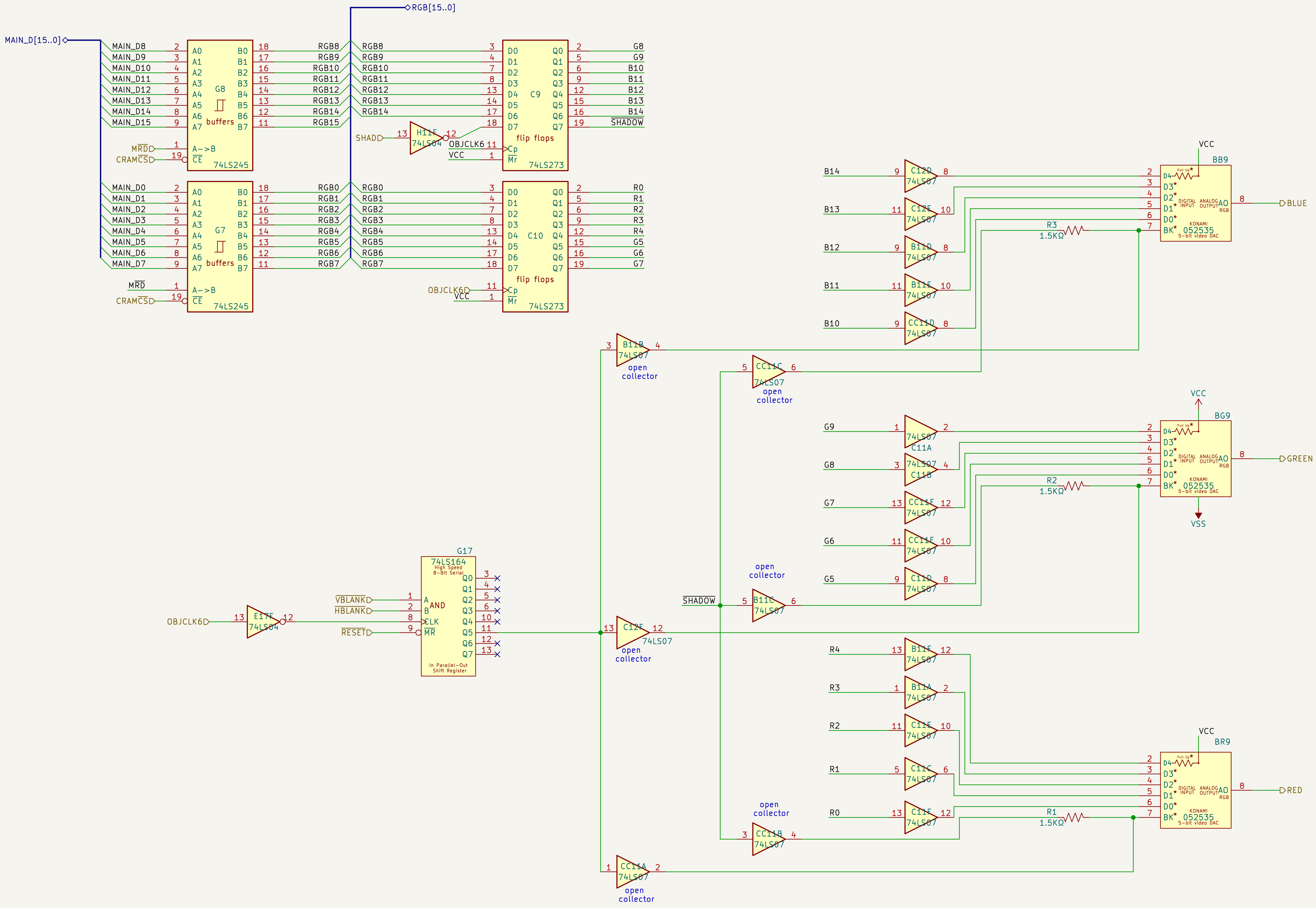
KONAMI PWB353018A GX065
Sheet: /colmix/
File: colmix.kicad_sch

Title: X-Men

Size: A3
Date: 2023-07-17
Rev: José Tejada
KiCad E.D.A. 8.0.5
Id: 6/10


MAIN_D[15..0]
RGB[15..0]
MAIN_D8
MAIN_D9
MAIN_D10
MAIN_D11
MAIN_D12
MAIN_D13
MAIN_D14
MAIN_D15
G8
buffers
74LS245
A->B
CE
MRD
CRAMCS
RGB8
RGB9
RGB10
RGB11
RGB12
RGB13
RGB14
RGB15
C9
flip flops
74LS273
Cp
Mr
G8
G9
B10
B11
B12
B13
B14
SHADOW
SHADD
H11F
74LS04
OBJCLK6
VCC
MAIN_D0
MAIN_D1
MAIN_D2
MAIN_D3
MAIN_D4
MAIN_D5
MAIN_D6
MAIN_D7
G7
RGB0
RGB1
RGB2
RGB3
RGB4
RGB5
RGB6
RGB7
C10
R0
R1
R2
R3
R4
G5
G6
G7
B14
B13
B12
B11
B10
C12D
C12E
B11D
B11E
CC11D
74LS07
VCC
BB9
R3
1.5KΩ
DIGITAL INPUT
ANALOG OUTPUT
KONAMI
052535
5-bit video DAC
RGB
D4
D3
D2
D1
D0
BK
AO
BLUE
B11B
open collector
CC11C
G9
G8
G7
G6
G5
C11A
C11B
CC11E
CC11F
C11D
B11C
BG9
R2
GREEN
VSS
G17
74LS164
High Speed 8-Bit Serial
In Parallel-Out Shift Register
VBLANK
HBLANK
RESET
AND
CLK
MR
Q0
Q1
Q2
Q3
Q4
Q5
Q6
Q7
OBJCLK6
E17F
C12F
R4
R3
R2
R1
R0
B11F
B11A
C11E
C11C
C11F
BR9
R1
RED
CC11B
CC11A

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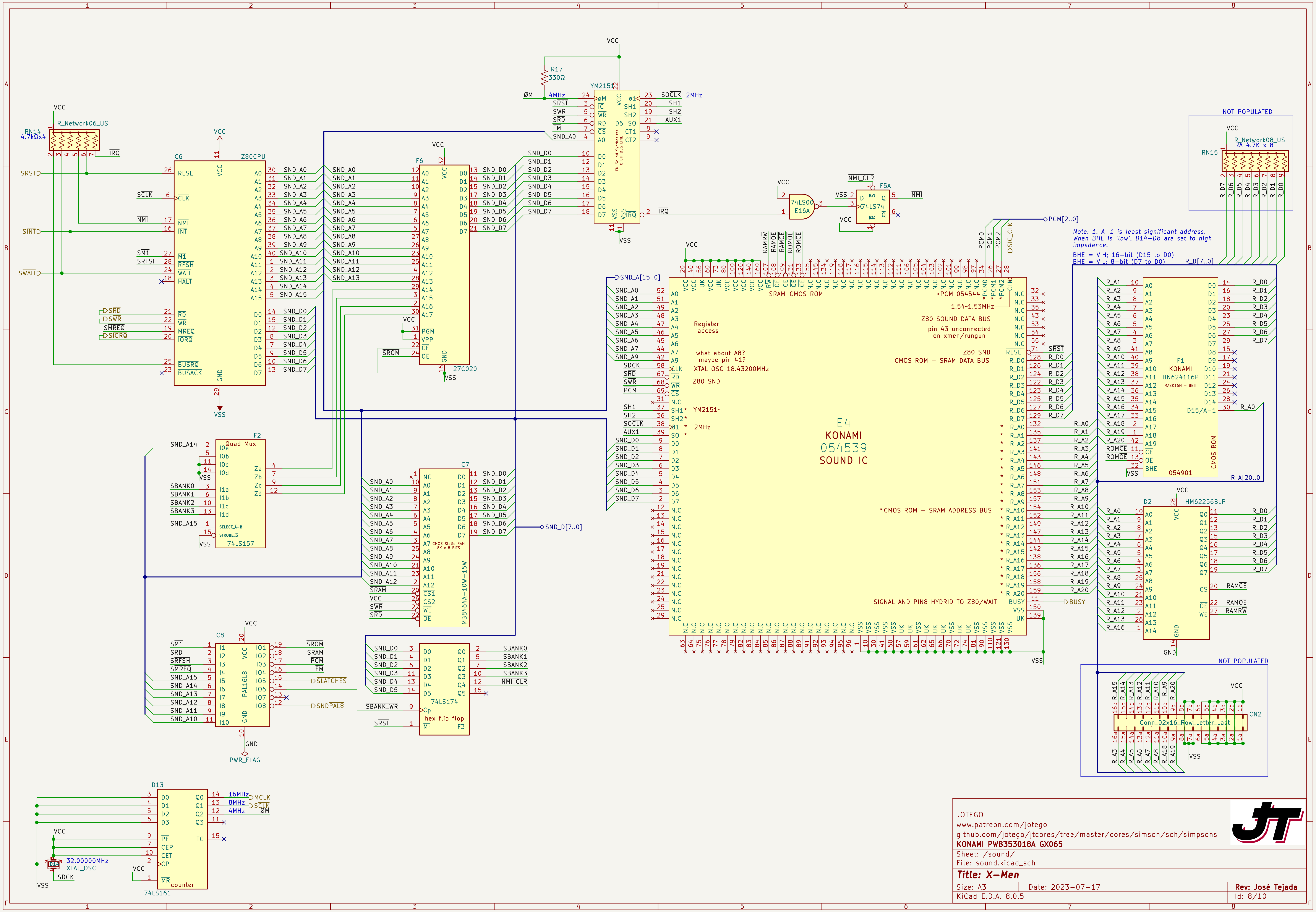
KONAMI PWB353018A GX065

Sheet: /colmix/rgb/
File: rgb.kicad_sch

Title: X-Men

Size: A3 Date: 2023-07-17 **Rev: José Tejada**
KiCad E.D.A. 8.0.5 Id: 7/10


VCC
R17
330Ω
YM2151
ØM
4MHz
SRST
SWR
SRD
FM
SND_A0
ø1
SOCLK
2MHz
SH1
SH2
AUX1
IC
WR
RD
CS
A0
D6 SO
CT1
CT2
FM Sound Synthesizer 8 BIT BUS LINE
IRQ
VSS
RN14
4.7kΩx4
R_Network06_US
C6
Z80CPU
RESET
CLK
NMI
INT
M1
RFSH
WAIT
HALT
RD
WR
MREQ
IORQ
BUSREQ
BUSACK
GND
SRST
SCLK
SINT
SWAIT
SM1
SRFSH
SRD
SWR
SMREQ
SIORQ
F6
27C020
PGM
VPP
CE
OE
SROM
SND_A0 … SND_A15
SND_D0 … SND_D7
74LS00
E16A
NMI_CLR
F5A
74LS74
NMI
SND_A[15..0]
PCM[2..0]
PCM0
PCM1
PCM2
SIC_CLK
RAMRW
RAMOE
RAMCE
ROMOE
ROMCE
SRAM CMOS ROM
*PCM 054544
1.54-1.53MHz
Z80 SOUND DATA BUS
pin 43 unconnected
on xmen/rungun
Register
access
what about A8?
maybe pin 41?
XTAL OSC 18.43200MHz
Z80 SND
YM2151*
2MHz
Z80 SND
CMOS ROM - SRAM DATA BUS
E4
KONAMI
054539
SOUND IC
*CMOS ROM - SRAM ADDRESS BUS
SIGNAL AND PIN8 HYDRID TO Z80/WAIT
BUSY
VSS
UK
N.C
R_A0 … R_A20
R_D0 … R_D7
NOT POPULATED
RN15
R_Network08_US
RA 4.7K x 8
Note: 1. A-1 is least significant address.
When BHE is 'low', D14-D8 are set to high impedance.
BHE = VIH; 16-bit (D15 to D0)
BHE = VIL; 8-bit (D7 to D0)
R_D[7..0]
F1
KONAMI
HN624116P
MASK16M - 8BIT
CMOS ROM
054901
ROMCE
ROMOE
BHE
D15/A-1
R_A[20..0]
D2
HM62256BLP
RAMCE
RAMOE
RAMRW
GND
F2
Quad Mux
74LS157
SELECT_Ā-B
STROBE_Ē
SBANK0
SBANK1
SBANK2
SBANK3
SND_A14
SND_A15
C7
CMOS Static RAM 8K x 8 BITS
MB8464A-10W-15W
SRAM
CS1
CS2
WE
SND_D[7..0]
C8
PAL16L8
SROM
SRAM
PCM
FM
SLATCHES
SNDPAL8
PWR_FLAG
F3
74LS174
hex flip flop
SBANK_WR
SBANK0
SBANK1
SBANK2
SBANK3
NMI_CLR
NOT POPULATED
CN2
Conn_02x16_Row_Letter_Last
D13
74LS161
counter
16MHz
8MHz
4MHz
MCLK
SCLK
ØM
D14
32.00000MHz
XTAL_OSC
SDCK
PE
CEP
CET
CP
MR
TC

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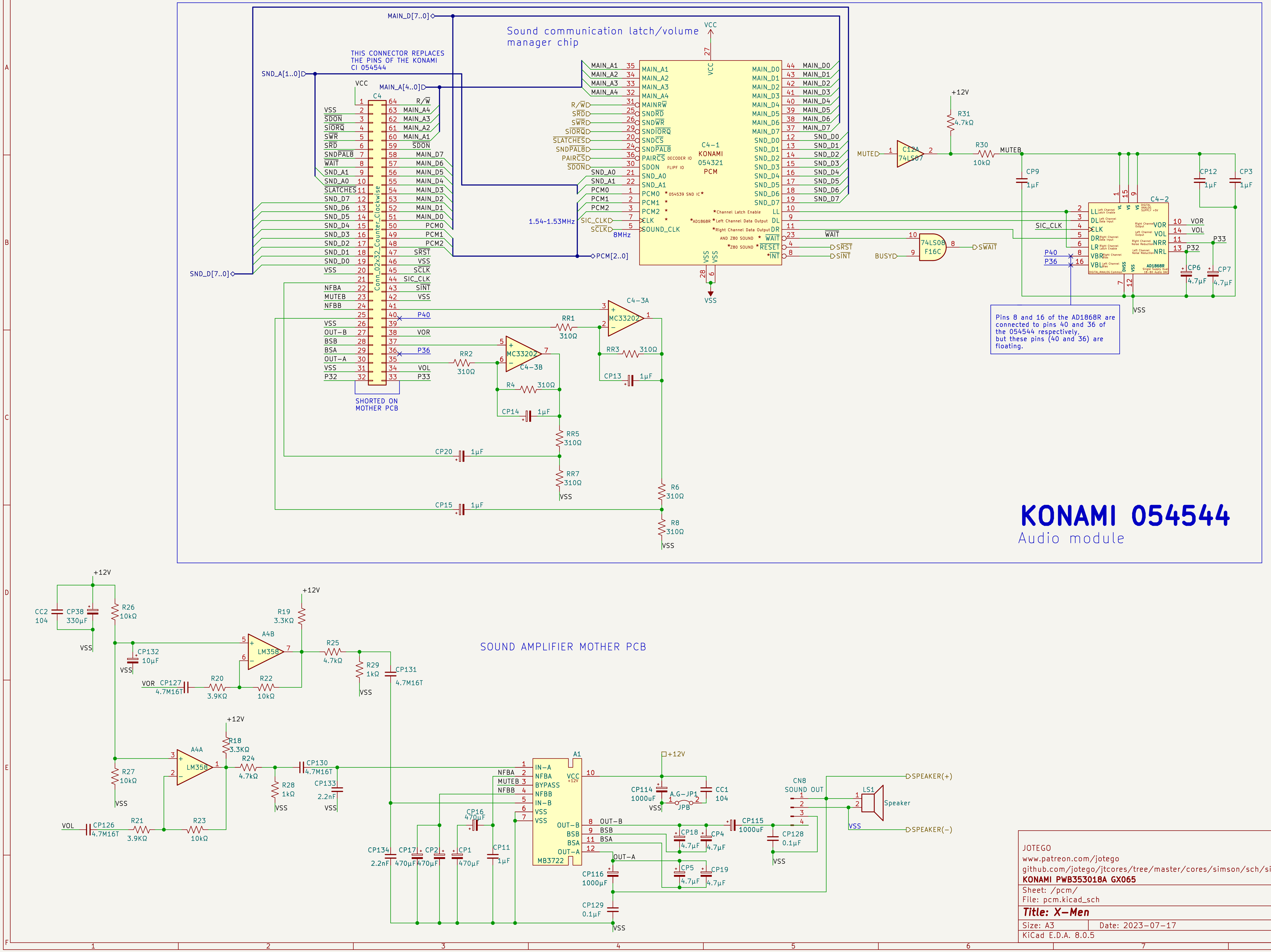
KONAMI PWB353018A GX065

Sheet: /sound/
File: sound.kicad_sch

Title: X-Men

Size: A3	Date: 2023-07-17	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 8/10


Sound communication latch/volume manager chip
THIS CONNECTOR REPLACES THE PINS OF THE KONAMI CI 054544
MAIN_D[7..0]
SND_A[1..0]
MAIN_A[4..0]
SND_D[7..0]
PCM[2..0]
VCC
VSS
+12V
C4
Conn_02x32_Counter_Clockwise
SHORTED ON MOTHER PCB
C4-1
KONAMI
054321
PCM
054539 SND IC
DECODER IO
FLIPF IO
AD1868R
Channel Latch Enable
Left Channel Data Output
Right Channel Data Output
AND Z80 SOUND
Z80 SOUND
1.54-1.53MHz
8MHz
SIC_CLK
SCLK
MUTED
C12A
74LS07
R31
4.7kΩ
R30
10kΩ
MUTEB
CP9
CP12
CP3
1µF
C4-2
AD1868R
Single Supply Dual 18-Bit Audio DAC
VOR
VOL
P33
P32
P40
P36
CP6
CP7
4.7µF
74LS08
F16C
BUSY
SWAIT
SRST
SINT
WAIT
Pins 8 and 16 of the AD1868R are connected to pins 40 and 36 of the 054544 respectively, but these pins (40 and 36) are floating.
C4-3A
C4-3B
MC33202
RR1
RR2
RR3
RR5
RR7
310Ω
R4
R6
R8
CP13
CP14
CP15
CP20
KONAMI 054544
Audio module
SOUND AMPLIFIER MOTHER PCB
CC2
104
CP38
330µF
R26
10kΩ
CP132
10µF
A4B
LM358
R19
3.3KΩ
R25
4.7kΩ
R29
1kΩ
CP131
4.7M16T
CP127
4.7M16T
R20
3.9KΩ
R22
10kΩ
A4A
R18
3.3KΩ
R24
4.7kΩ
R27
10kΩ
R28
1kΩ
CP130
4.7M16T
CP133
2.2nF
CP126
4.7M16T
R21
3.9KΩ
R23
10kΩ
A1
MB3722
IN-A
NFBA
BYPASS
NFBB
IN-B
OUT-B
BSB
BSA
OUT-A
CP134
2.2nF
CP17
470µF
CP2
470µF
CP16
470µF
CP1
470µF
CP11
1µF
CP114
1000µF
A.G-JP1
JPB
CC1
104
CP115
1000µF
CP128
0.1µF
CP18
CP4
CP5
CP19
4.7µF
CP116
1000µF
CP129
0.1µF
CN8
SOUND OUT
LS1
Speaker
SPEAKER(+)
SPEAKER(-)

JOTEGO
www.patreon.com/jotego
github.com/jotego/jtcores/tree/master/cores/simson/sch/simpsons
KONAMI PWB353018A GX065

Sheet: /pcm/
File: pcm.kicad_sch

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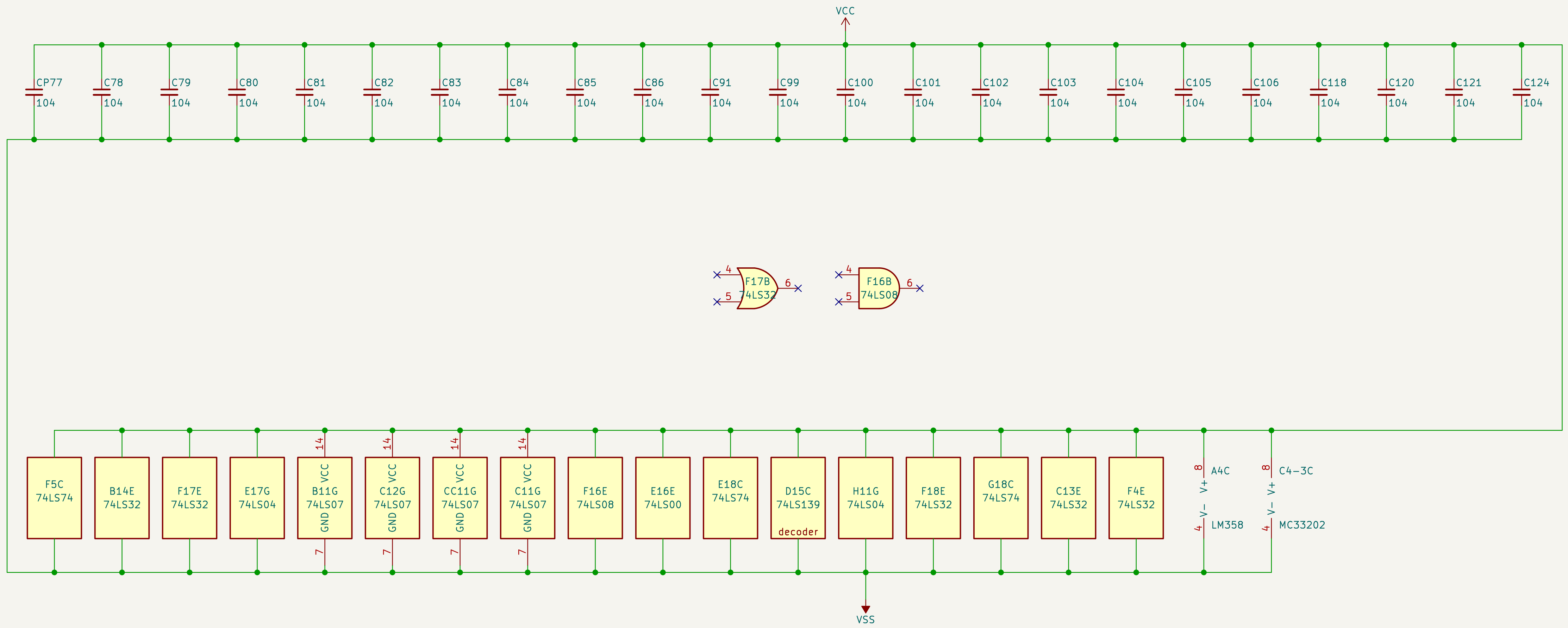
VCC
CP77 104
C78 104
C79 104
C80 104
C81 104
C82 104
C83 104
C84 104
C85 104
C86 104
C91 104
C99 104
C100 104
C101 104
C102 104
C103 104
C104 104
C105 104
C106 104
C118 104
C120 104
C121 104
C124 104
4
5
F17B
74LS32
6
4
5
F16B
74LS08
6
F5C 74LS74
B14E 74LS32
F17E 74LS32
E17G 74LS04
14 VCC B11G 74LS07 GND 7
14 VCC C12G 74LS07 GND 7
14 VCC CC11G 74LS07 GND 7
14 VCC C11G 74LS07 GND 7
F16E 74LS08
E16E 74LS00
E18C 74LS74
D15C 74LS139 decoder
H11G 74LS04
F18E 74LS32
G18C 74LS74
C13E 74LS32
F4E 74LS32
8 V+ A4C V− 4 LM358
8 V+ C4-3C V− 4 MC33202
VSS

JOTEGO
www.patreon.com/jotego
github.com/jotego/jtcores/tree/master/cores/simson/sch/simpsons
KONAMI PWB353018A GX065
Sheet: /capacitors/
File: capacitors.kicad_sch
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Size: A3 Date: 2023-07-17 **Rev: José Tejada**
KiCad E.D.A. 8.0.5 Id: 9/10